

Classroom Problem-Solving Activity*Classes & Methods: The Student Report Card Builder*

Course topic	Classes, Objects, and Methods in Java
Concepts practiced	Scanner input, method arguments, void vs non-void methods, method overloading, method reuse, class attributes, access modifiers, accessor & mutator methods, toString(), passing objects as parameters
Format	Group work, in-class coding activity
Duration	40–45 minutes

1. Problem Statement

A teacher wants a small console program to build report cards for two students. For each student, the program asks for a name and two test marks using Scanner, builds a Student object from that input, and prints a report. The teacher also wants a quick way to compare two students to see who has the higher average, and a way to add bonus marks with or without a reason.

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Enter name for student 1: Taylor
Enter mark 1: 78
Enter mark 2: 85

Enter name for student 2: Jereme
Enter mark 1: 90
Enter mark 2: 88

Taylor -> marks: 78, 85, average: 81.5
Jereme -> marks: 90, 88, average: 89.0

Jereme has a higher average than Taylor.

Taylor receives 5 bonus marks.
Jereme receives 3 bonus marks for: Class participation.
```

2. What You Need to Build

1. A Student class with three private attributes: name (String), mark1 (int), mark2 (int).
2. A constructor Student(String name, int mark1, int mark2) that sets all three attributes.
3. A second, overloaded constructor Student(String name) that reuses the first one — call it with this(name, 0, 0).
4. Accessor (non-void) methods: getName(), getMark1(), getMark2(), and getAverage(), where getAverage() returns (mark1 + mark2) / 2.0.
5. Mutator (void) methods: setMark1(int mark1) and setMark2(int mark2).
6. An overridden toString() method that returns a line such as: name + " -> marks: " + mark1 + ", " + mark2 + ", average: " + getAverage() .
7. A method hasHigherAverage(Student other) that takes another Student object as-a parameter and returns true if this student's average is higher than the other student's average.
8. An overloaded pair of void methods: addBonus(int bonus), which adds bonus to mark1 and prints a message, and addBonus(int bonus, String reason), which reuses addBonus(int bonus) and then prints the reason as well.
9. A main method that uses Scanner to read a name and two marks for two students, builds two Student objects using the arguments read from the keyboard, prints both using toString(), calls hasHigherAverage(...) to report who scored higher, and calls both versions of addBonus(...) — one with a reason and one without.